Mateusz Jaszczuk

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Education

University of Pennsylvania *Philadelphia, PA*
Ph.D. in Mechanical Engineering and Applied Mechanics *Aug 2026 – Present*
GPA: 4.00/4.00

University of Pennsylvania *Philadelphia, PA*
M.S.E. in Mechanical Engineering and Applied Mechanics – Mechatronics and Robotics *Aug 2024 – May 2026*
GPA: 4.00/4.00
Thesis: *Learning to Feel: Force-Aware Data-Driven Estimation and Control for Adaptive Physical Interactions*

Purdue University *West Lafayette, IN*
B.S. in Aeronautical and Astronautical Engineering – Autonomy and Control *Aug 2020 – May 2024*
GPA: 3.71/4.00

Skills

Programming & Tools - Python (PyTorch, JAX), C++, Isaac Lab, MuJoCo, MATLAB, ROS, PyBullet, Drake, CUDA

Machine Learning - Reinforcement Learning (PPO, SAC, Q-Learning), Imitation Learning, Behavior Cloning, Diffusion, Sim-to-Real (Domain Randomization), Neural Networks (MLP, CNN, Transformers, NeRF), Probabilistic Modeling

Control & Estimation - Impedance Control, Operational Space Control, Inverse Dynamics, Optimal Control (MPC, iLQR, DP), State Estimation (EKF, UKF, PF), Safety-Critical Control (CLF-QP, CBF-QP), System Identification, SLAM

Hardware - Franka Research 3 (7-DoF), LiDAR, RealSense RGB-D, OptiTrack Motion Capture, Sensor Integration

Experience

Figueroa Robotics Lab (GRASP) *Philadelphia, PA*
Graduate Research Assistant, Ph.D. *Aug 2026 – Present*

- Developing a policy-agnostic framework that estimates user intent online from language instructions and physical interaction cues, enabling a meta-policy to make existing frozen robot policies interactive by responding to ambiguity, adapting to changing goals, and asking clarifying questions during deployment.
- Developing a sim-to-real reinforcement learning framework to compensate for unknown payload, actuator, and friction dynamics, supporting reliable policy deployment and precise, compliant control under changing conditions.

Graduate Research Assistant, M.S.E. *Jan 2025 – Aug 2026*

- Designed a transformer-based adaptive controller that fuses point cloud and force data to distinguish human perturbations from payload changes in real time, achieving simultaneous unknown-load compensation, precise tracking performance, and compliant response to physical interactions; reported in an arXiv preprint [1].
- Developed *Rapid Mismatch Estimation* – a real-time data-driven framework identifying unknown payload dynamics on a 7 DoF manipulator, enabling safe and reactive manipulation under load uncertainty; published at CoRL 2025 [2].
- Supported research on learning DS from demonstrations, contributing to imitation learning-based policy development.
- Implemented and deployed a DS-based operational-space impedance controller on a Franka Research 3 using custom ROS packages, enabling compliant and reactive manipulation; validated in simulation and on hardware.

Composite Manufacturing & Simulation Center *West Lafayette, IN*
Undergraduate Research Assistant *Oct 2022 – May 2024*

- Developed a Physics-Guided Transfer Learning framework using Bayesian model with Gaussian Process surrogates for data-efficient composite's property prediction across varying conditions, reducing required characterization efforts.
- Co-authored a peer-reviewed journal publication and designed visualizations to demonstrate the model's evaluation.

Air Force Research Lab - Purdue UAS Research and Test Facility *West Lafayette, IN*
Undergraduate Research Assistant, Project Lead *Aug 2023 – Dec 2023*

- Developed autonomous fixed-wing aircraft for enclosed hangar-flight with Windracers (UK), including sizing, flight simulations, and led motion capture integration for indoor localization and autonomous safety protocols validation.
- Managed project coordination and communication, aligning the development cycle with the client's requirements.

Publications and Presentations

[1] **M. Jaszczuk**, P. Perdikaris, and N. Figueroa, "Multimodal interaction estimation for contact-aware adaptive impedance control." arXiv:2610.24235 [cs.RO], 2026.

[2] **M. Jaszczuk** and N. Figueroa, "Rapid mismatch estimation via neural network informed variational inference," in *9th Annual Conference on Robot Learning (CoRL)*, 2025.

[3] A. J. Thomas, **M. Jaszczuk**, E. Barocio, G. Ghosh, I. Bilionis, and R. B. Pipes, "Probabilistic physics-guided transfer learning for material property prediction in extrusion deposition additive manufacturing," *Computer Methods in Applied Mechanics and Engineering*, vol. 419, p. 116660, 2024.

[4] A. J. Thomas, **M. Jaszczuk**, E. Barocio, I. Bilionis, and R. B. Pipes, "Physics-guided transfer learning for property prediction in composite additive manufacturing." SAMPE Japan, 2023.

[5] **M. Jaszczuk**, "A bayesian framework for transfer of process-dependent material properties." Purdue Fall Undergraduate Research Expo, November 2023. West Lafayette, IN, USA.

Awards

Outstanding Research Award *Philadelphia, PA*
Awarded by Department of Mechanical Engineering, University of Pennsylvania *May 2026*

Project Experience

Small-Scale Transformer for Text Generation - Graduate Course Project *Philadelphia, PA*
Student Researcher, Developer *Aug 2024 – Dec 2024*

- Implemented a lightweight GPT-inspired transformer neural network to generate coherent artificial text.
- Designed multi-head attention, positional encoding, and MLP modules in PyTorch, trained the model on small datasets (books, screenplays) to demonstrate artificial text generation and performance evaluation.
- Conducted ablation on regularization (dropout, layer normalization) to evaluate the impact on performance.

Autonomous Mobile Battle Robot - Graduate Course Project *Philadelphia, PA*
Robotics Engineer, Team Member *Aug 2024 – Dec 2024*

- Designed a mobile robot for autonomous competition task execution (wall-following, navigation, attack) across a full project lifecycle: concept design, prototyping, integration, testing, and final competition.
- Integrated mechanical and electronic subsystems (platform, motors, ultrasonic/ToF sensors, ESP32 microcontroller, custom circuits) and programmed an Arduino-based embedded system for real-time control.

NOAA Wind Tracking Satellite Constellation - Senior Design Project *West Lafayette, IN*
Attitude Dynamics and Control Engineer, Team Member *Aug 2023 – Dec 2023*

- Developed a MATLAB 6-DOF model to simulate satellite attitude dynamics across multiple orbital planes.
- Coordinated system integration and designed active/passive attitude control strategies, ensuring the spacecraft met stability and maneuverability requirements for long-duration Earth's surface monitoring.

Purdue Space Program NASA Student Launch *West Lafayette, IN*
Project Manager (2022-2023), Launch Vehicle Construction Engineer (2021-2022) *Aug 2021 – May 2023*

- Led a 60-member interdisciplinary team through the full project life cycle for the NASA Student Launch competition.
- Coordinated project deliverables and communications with NASA, ensuring the system meets mission requirements.
- Led launch vehicle development and 6-DOF flight simulations for trajectory optimization and mission design.

Teaching Experience

University of Pennsylvania *Philadelphia, PA*
MEAM 5200 Introduction to Robotics - *Teaching Assistant* *Fall 2025*

Professional Service

IEEE International Conference on Robotics and Automation (ICRA) – Reviewer
IEEE Transactions on Robotics (T-RO) – Reviewer
Conference on Robot Learning (CoRL) – Reviewer